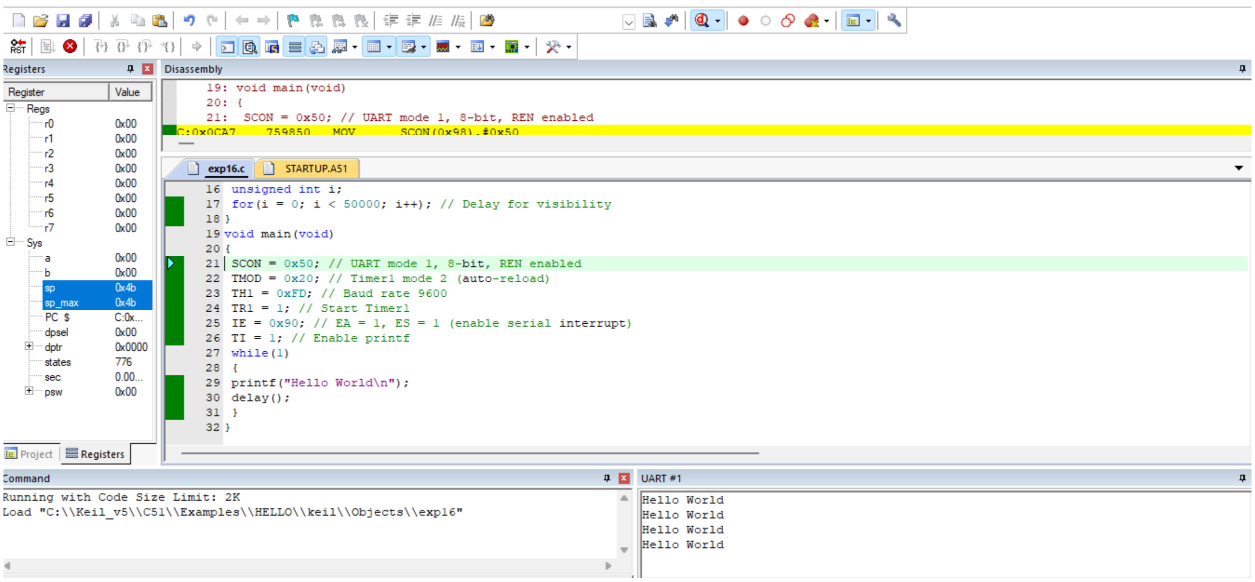
Embedded syatems exp-16

Program:

```
#include <reg51.h>
#include <stdio.h>
void serial_ISR(void) interrupt 4
{
 char ch;
 if (RI)
 {
 RI = 0; // Clear receive interrupt flag
 ch = SBUF; // Read received character
 printf("\n>>> INTERRUPT RECEIVED: ");
 printf("%c\n", ch);
 }
}
void delay(void)
{
 unsigned int i;
 for(i = 0; i < 50000; i++); // Delay for visibility
}
void main(void)
{
 SCON = 0x50; // UART mode 1, 8-bit, REN enabled
 TMOD = 0x20; // Timer1 mode 2 (auto-reload)
 TH1 = 0xFD; // Baud rate 9600
 TR1 = 1; // Start Timer1
 IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
 TI = 1; // Enable printf
 while(1)
```

```
{
printf("Hello World\n");
delay();
}
}
```

Output:



When interrupt received